OPTIONS PRICING

Option pricing models aim to estimate the fair value of options based on underlying asset dynamics and market conditions. In this report, we analyze real market option data and study option behavior using the Black–Scholes model, Greeks analysis, implied volatility smile, Binomial model, and Monte Carlo simulation. Market prices are compared with theoretical prices to evaluate model accuracy. The study highlights how volatility behavior and model assumptions influence pricing. By combining analytical and simulation-based approaches, the report provides a comprehensive view of option valuation and risk sensitivity

1.Greeks Analysis

1.1 Introduction

In options pricing, **Greeks** measure the sensitivity of option prices to changes in various parameters:

- **Delta (Δ):** Sensitivity to changes in the underlying asset price
- **Gamma (Γ):** Sensitivity of Delta to changes in the underlying price
- **Vega (ν):** Sensitivity to changes in volatility
- **Theta (Θ):** Sensitivity to time decay
- **Rho (ρ):** Sensitivity to changes in the risk-free interest rate

We calculated Greeks for both **Call** and **Put options** using the **Black-Scholes formulas** with the respective implied volatilities (`Call_IV` and `Put_IV`).

1.2 Methodology

1. **Data Preparation:** Added columns for each Greek:

   Calls: `call_delta`, `call_gamma`, `call_vega`, `call_theta`, `call_rho`
   Puts: `put_delta`, `put_gamma`, `put_vega`, `put_theta`, `put_rho`

2. **Greeks Calculation:**

For each strike, Greeks were computed using Black-Scholes formulas.
Inputs: `S = 1533.40`, Strike K, Implied Volatility σ, Risk-free rate `r`, Time to expiry T.
Results stored in the **DataFrame (colab).**

3. **Visualization:** Plotted each Greek against Strike Price to study sensitivity and behavior.

1.3 Observations from Plots

Call Greeks

- **Delta:** Decreases with strike. ITM ≈ 1, ATM ≈ 0.5, OTM ≈ 0. Delta shows how Call price moves with underlying.
- **Gamma:** Peaks near ATM (~1500). Maximum sensitivity of Delta to underlying changes; crucial for hedging.
- **Vega:** Highest near ATM, decreases for ITM/OTM. Indicates sensitivity to volatility; ATM options are most affected.
- **Theta:** Negative for all strikes, lowest near ATM. Shows time decay; ATM options lose value fastest.
- **Rho:** Positive, decreases with strike. Higher strike Calls are less sensitive to interest rates.

Put Greeks

- **Delta:** Negative, decreases with strike. ITM ≈ -1, ATM ≈ -0.5, OTM ≈ 0. Puts gain when underlying falls.
- **Gamma:** Peaks near ATM; ATM Puts have maximum Delta sensitivity.
- **Vega:** Peaks near ATM, similar to Calls. ATM Puts are most sensitive to volatility changes.
- **Theta:** Negative, smallest near ATM. ATM Puts experience fastest time decay.
- **Rho:** Negative, decreases with strike. Higher strike Puts lose more value as interest rates rise.

1.4 Summary

- Greeks provide essential insights into option sensitivity and **hedging strategies**.

- **ATM options** are most sensitive to underlying price (Gamma), volatility (Vega), and time decay (Theta).

- **ITM and OTM options** show extreme Delta values (±1 or 0) but lower sensitivity to volatility.

- Visualizing Greeks versus strike allows traders to **identify risk exposures and optimize hedging**.

2. Black-Scholes Option Pricing Model

2.1 Introduction

The Black-Scholes (BS) model provides theoretical prices for European call and put options using the underlying asset price (S), strike price (K), risk-free rate (r), time to maturity (T), and volatility (σ). Comparing BS prices with actual market prices helps identify mispricing and assess model accuracy.

2.2 Methodology

Black-Scholes Option Pricing Formulas

Call Option (C):
$C = S \times N(d1) - K \times e^{(-rT)} \times N(d2)$

Put Option (P):
$P = K \times e^{(-rT)} \times N(-d2) - S \times N(-d1)$

Where:
$d1 = [\ln(S / K) + (r + (\sigma^2 / 2)) \times T] / (\sigma \times \sqrt{T})$
$d2 = d1 - \sigma \times \sqrt{T}$

Parameters:

S = Current stock price
K = Strike price
r = Risk-free interest rate
T = Time to maturity (in years)
σ = Volatility of the underlying asset
N() = Cumulative standard normal distribution

For each strike:

Computed BS theoretical prices (Call & Put)
Calculated absolute and percentage differences from market prices

2.3 Results & Observations

- **Figure 1 – Call Options (Market vs BS):** Near-the-money calls closely follow BS prices. High-strike calls show increasing deviation.
- **Figure 2 – Put Options (Market vs BS):** Lower-strike puts align with BS values; deep ITM puts deviate due to market conditions.
- **Figure 3 – Call Mispricing:** Absolute and percentage differences highlight where calls are under- or overvalued. Deviations grow for OTM calls.
- **Figure 4 – Put Mispricing:** Shows relative mispricing; extreme ITM puts deviate more, indicating liquidity or skew effects.
- **Figure 5 – Combined View:** Calls and puts together reveal overall fit; near-the-money options consistently match BS predictions.

Key Observations:

- Near-the-money options fit BS model best.
- OTM and deep ITM options show larger discrepancies.
- Call deviations increase at high strikes; put deviations increase at low strikes.

2.4 Analysis & Conclusion

The BS model is a reliable benchmark for pricing European options. Deviations at extreme strikes are likely due to volatility skew, liquidity constraints, and market demand-supply factors. Comparing theoretical and market prices highlights mispriced strikes, assisting in trading, hedging, and risk assessment. For extreme strikes, alternative models may improve accuracy.

3. Implied Volatility Smile

3.1 Introduction

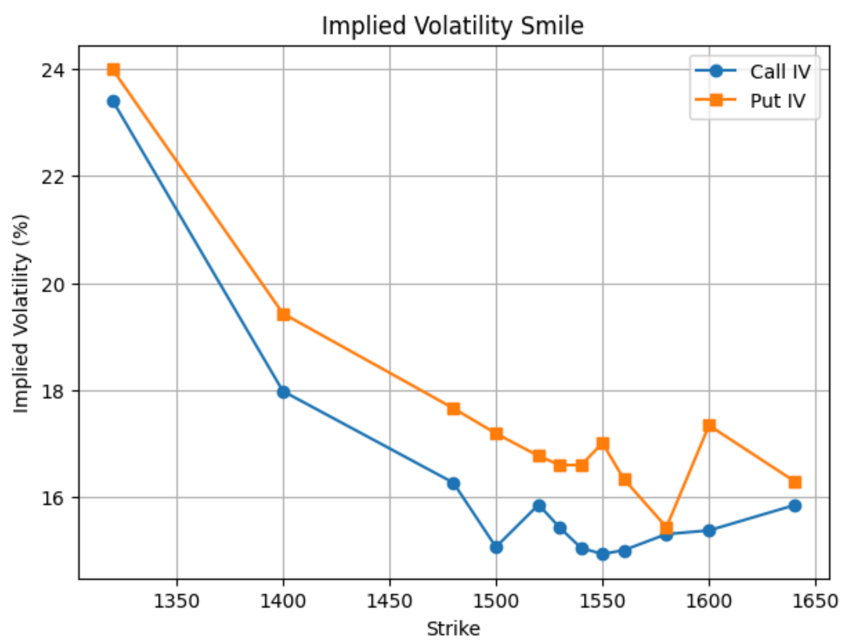
Implied volatility (IV) is the volatility value that, when used in an option pricing model, reproduces the observed market price of an option. Unlike historical volatility, IV reflects the market's expectation of future uncertainty.

In practice, implied volatility is not constant across strike prices, leading to a phenomenon known as the **volatility smile**.

3.2 Methodology

- Implied volatility values for calls and puts are extracted directly from market data.
- IV is plotted against strike prices for both call and put options.
- Separate and combined IV smile plots are analyzed.

3.3 Observations



- Implied volatility is lowest near at-the-money (ATM) strikes.
- IV increases for deep in-the-money (ITM) and out-of-the-money (OTM) options.
- Both call and put options exhibit a curved, smile-like pattern.

This confirms the presence of a volatility smile in real market data.

3.4 Conclusion

The volatility smile indicates that the market prices extreme price movements more aggressively. This behavior violates the constant volatility assumption used in classical pricing models and explains systematic deviations between model prices and market prices.

4. Binomial Option Pricing Model

4.1 Introduction

The Binomial Option Pricing Model is a discrete-time model that represents the evolution of the underlying asset price through a recombining tree of possible future prices. It is based on the principle of no-arbitrage and risk-neutral valuation.

4.2 Methodology

- The stock price is assumed to move up or down at each time step.
- Risk-neutral probabilities are computed using the risk-free rate.
- Option payoffs are calculated at maturity.
- Backward induction is used to compute the option price at the initial node.
- Strike-specific implied volatility is used as the volatility input.

Both call and put options are priced using this approach.

4.3 Observations

- Binomial prices follow the same trend as market prices across strikes.
- Prices decrease with increasing strike for calls and increase for puts.
- The model underprices options compared to market LTP, especially for deep ITM and OTM strikes.
- Pricing errors are smallest near ATM options.

4.4 Conclusion

The Binomial model captures the general structure of option prices but fails to fully match market prices due to simplifying assumptions such as constant volatility and frictionless markets. The presence of volatility smile contributes significantly to pricing errors.

5. Monte Carlo Simulation Model

5.1 Introduction

Monte Carlo simulation is a probabilistic method that estimates option prices by simulating a large number of possible future paths of the underlying asset price. It is particularly useful for complex derivatives and provides flexibility in modeling.

5.2 Methodology

- Stock price paths are simulated using Geometric Brownian Motion.
- Risk-neutral drift and implied volatility are used.
- Option payoff is computed at maturity for each path.
- The discounted average payoff gives the option price.
- A large number of simulations ensures convergence.

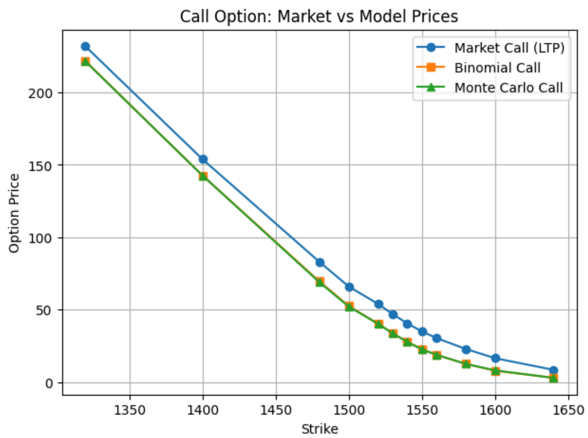
Both call and put options are priced using Monte Carlo simulation.

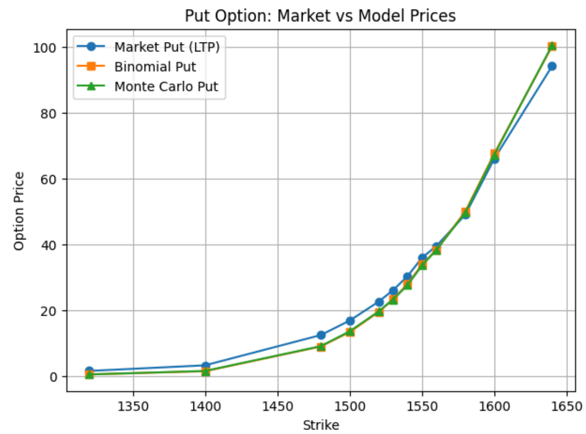
5.3 Observations

- Monte Carlo prices closely match Binomial model prices.
- The trend of market prices is well captured.
- Slight randomness is observed due to simulation-based estimation.
- Pricing deviations increase for far-from-ATM strikes.

5.4 Conclusion

Monte Carlo simulation provides reliable theoretical prices and converges to Binomial model values. However, like the Binomial model, it is limited by constant volatility assumptions and does not fully capture market risk premiums.





Observations:

- Market prices are consistently higher than model prices.
- Models perform best near ATM strikes.
- Errors increase for deep ITM and OTM options.
- Binomial and Monte Carlo errors are similar in magnitude.

Conclusion:

This study shows that the Black–Scholes model serves as a strong analytical benchmark, particularly for near-the-money options, while Greeks effectively explain option sensitivity to price, volatility, time, and interest rates. The presence of an implied volatility smile confirms that volatility is strike-dependent, violating constant-volatility assumptions. Binomial and Monte Carlo models successfully capture the overall pricing trend and produce consistent theoretical prices. However, both models systematically underprice options compared to market LTP, especially for deep ITM and OTM strikes. Overall, the analysis highlights the gap between idealized pricing models and real market behavior, emphasizing the need for advanced volatility modeling.